Chapter 15 Binomial Theorem - Homework Answers

Chapter revision | Warm-up

1
a $\dfrac{7!}{5!} = 7 \times 6 = 42$
b ${}^9C_3 = \dfrac{9 \times 8 \times 7}{3 \times 2 \times 1} = 84$
c ${}^{12}C_{10} = {}^{12}C_2 = \dfrac{12 \times 11}{2 \times 1} = 66$
d ${}^{10}C_0 = {}^{10}C_{10} = 1$ and ${}^{10}C_1 = {}^{10}C_9 = 10$, so the sum is $1 + 10 + 10 + 1 = 22$.

2
a ${}^nC_1 = \dfrac{n!}{1!\,(n-1)!} = n$ and ${}^nC_2 = \dfrac{n!}{2!\,(n-2)!} = \dfrac{n(n-1)}{2}$
b $\dfrac{n(n-1)}{2} = 45$ gives $n^2 - n - 90 = 0$, so $(n-10)(n+9) = 0$. Since $n > 0$, $n = 10$.

3
a $(2x^2)^3 = 2^3x^6 = 8x^6$
b $\dfrac{9}{x^2} \times x^3 = 9x$
c $\left(-\dfrac{2}{x}\right)^3 = \dfrac{(-2)^3}{x^3} = -\dfrac{8}{x^3}$
d $8^{-2/3} = \dfrac{1}{8^{2/3}} = \dfrac{1}{(\sqrt[3]{8})^2} = \dfrac{1}{4}$

4
a $(2x-3)^2 = 4x^2 - 12x + 9$
b $(1+x)(1+2x+x^2) = 1 + 2x + x^2 + x + 2x^2 + x^3 = 1 + 3x + 3x^2 + x^3$
c The x^2 terms come from $1 \times x^2$ and $2x \times (-x)$, so the coefficient is $1 - 2 = -1$.

5
a $\dfrac{5!}{3!\,2!} = \dfrac{120}{12} = 10$ arrangements
b $10 = {}^5C_2$, the number of ways to choose which 2 of the 5 positions are the Bs. Choosing the 3 positions for the As gives ${}^5C_3$, which is also 10.

Chapter 15 Binomial Theorem - Homework Answers

Lesson 15A | Binomial expansions and Pascal's triangle

Lesson Examples

C1 **Tutor demo.**

a $(1+3a)^4 = 1+12a+54a^2+108a^3+81a^4$

b $(1-x)^5 = 1-5x+10x^2-10x^3+5x^4-x^5$

c $\left(1-\frac{1}{2}x\right)^5 = 1-\frac{5}{2}x+\frac{5}{2}x^2-\frac{5}{4}x^3+\frac{5}{16}x^4-\frac{1}{32}x^5$

Your turn.

a $(1+5a)^3 = 1+15a+75a^2+125a^3$

b $(1-x)^3 = 1-3x+3x^2-x^3$

c $\left(1-\frac{1}{3}x\right)^4 = 1-\frac{4}{3}x+\frac{2}{3}x^2-\frac{4}{27}x^3+\frac{1}{81}x^4$

C2 **Tutor demo.**

a $1+0.07+0.0021+0.000035+0.00000035+\cdots \doteq 1.07214$

b $1-0.05+0.001-0.00001+0.00000005-\cdots \doteq 0.95099$

Your turn.

a $1+0.12+0.006+0.00016+0.0000024+\cdots \doteq 1.12616$

b $1-0.08+0.0024-0.000032+0.00000016 \doteq 0.92237$

C3 **Tutor demo.**

a $\left(1+\frac{3}{x}\right)^2 = 1+6x^{-1}+9x^{-2}$ and $(1-x)^6 = 1-6x+15x^2-20x^3+\cdots$

b (i) $1-36+135=100$ (ii) $-6x+90x-180x=-96x$

Your turn.

a $\left(1+\frac{2}{x}\right)^2 = 1+4x^{-1}+4x^{-2}$ and $(1-x)^5 = 1-5x+10x^2-10x^3+\cdots$

b (i) $1-20+40=21$ (ii) $-5x+40x-40x=-5x$

C4

Tutor demo. a $80k^3x^3$ and $40k^2x^2$ b $\dfrac{80k^3}{40k^2} = 2k = \dfrac{3}{4}$, so $k = \frac{3}{8}$

Your turn. a $20k^3x^3$ and $15k^2x^2$ b $\dfrac{20k^3}{15k^2} = \dfrac{4}{3}k = 2$, so $k = \frac{3}{2}$

C5 **Tutor demo.** $\left(1+x+x^2\right)^3 = 1+3x+6x^2+7x^3+6x^4+3x^5+x^6$

Your turn. $\left(1+x-x^2\right)^3 = 1+3x-5x^3+3x^5-x^6$ (the terms in x^2 and x^4 cancel)

Textbook Homework

1 Coefficients of $(1+x)^n$:

n	x^0	x^1	x^2	x^3	x^4	x^5	x^6	x^7	x^8	x^9	x^{10}	x^{11}	x^{12}
0	1												
1	1	1											
2	1	2	1										
3	1	3	3	1									
4	1	4	6	4	1								
5	1	5	10	10	5	1							
6	1	6	15	20	15	6	1						
7	1	7	21	35	35	21	7	1					
8	1	8	28	56	70	56	28	8	1				
9	1	9	36	84	126	126	84	36	9	1			
10	1	10	45	120	210	252	210	120	45	10	1		
11	1	11	55	165	330	462	462	330	165	55	11	1	
12	1	12	66	220	495	792	924	792	495	220	66	12	1

2 a $1+6x+15x^2+20x^3+15x^4+6x^5+x^6$
b $1-6x+15x^2-20x^3+15x^4-6x^5+x^6$
c $1+9x+36x^2+84x^3+126x^4+126x^5+84x^6+36x^7+9x^8+x^9$
d $1-9x+36x^2-84x^3+126x^4-126x^5+84x^6-36x^7+9x^8-x^9$
e $1+5c+10c^2+10c^3+5c^4+c^5$
f $1+8y+24y^2+32y^3+16y^4$
g $1+\frac{7x}{3}+\frac{7x^2}{3}+\frac{35x^3}{27}+\frac{35x^4}{81}+\frac{7x^5}{81}+\frac{7x^6}{729}+\frac{x^7}{2187}$
h $1-9z+27z^2-27z^3$
i $1-\frac{8}{x}+\frac{28}{x^2}-\frac{56}{x^3}+\frac{70}{x^4}-\frac{56}{x^5}+\frac{28}{x^6}-\frac{8}{x^7}+\frac{1}{x^8}$
j $1+\frac{10}{x}+\frac{40}{x^2}+\frac{80}{x^3}+\frac{80}{x^4}+\frac{32}{x^5}$
k $1+\frac{5y}{x}+\frac{10y^2}{x^2}+\frac{10y^3}{x^3}+\frac{5y^4}{x^4}+\frac{y^5}{x^5}$
l $1+\frac{12x}{y}+\frac{54x^2}{y^2}+\frac{108x^3}{y^3}+\frac{81x^4}{y^4}$

4 a (i) $55x^2$ (ii) $165x^8$ b (i) $-35x^3$ (ii) $-21x^5$ c (i) $240x^4$ (ii) $192x^5$ d (i) $-\frac{12}{x}$ (ii) $\frac{54}{x^2}$

6 a $\big(1+(x-1)\big)^3=x^3$ b $\big(1-(x+1)\big)^6=(-x)^6=x^6$

7 21

8 a $a=76, b=44$ b $a=16, b=-8$

10 a 1.01814 b 0.81537

11 a (i) $1+4x+6x^2+\cdots$ (ii) -14
b (i) $1+10x+40x^2+80x^3+\cdots$ (ii) 40
c (i) $1-12x+54x^2-108x^3+\cdots$ (ii) -228

12 a $x=0$ or $\frac{1}{2}$ b $x=0, 1$ or 5

13 a -12 b 0 c 380 d $-\frac{5}{3}$

14 a 97 b $1\frac{10}{27}$

15 a (i) $15x^2$ (ii) $20x^3$ (iii) $3:4x$ (iv) $135, 540, 1:4$
b (i) $\frac{224}{81x^5}$ (ii) $\frac{448}{729x^6}$ (iii) $9x:2$ (iv) $\frac{7}{81}, \frac{7}{729}, 9:1$

16 a $k=5$ b $k=-2$ or 0

18 1.0634

19 $(1+x+y)^0=1$, $(1+x+y)^1=1+x+y$,
$(1+x+y)^2=1+2x+2y+2xy+x^2+y^2$,
$(1+x+y)^3=1+3x+3y+6xy+3x^2+3y^2+3x^2y+3xy^2+x^3+y^3$,
$(1+x+y)^4=1+4x+4y+12xy+6x^2+6y^2+6x^2y^2+12x^2y+12xy^2+4x^3+4y^3+4x^3y+4xy^3+x^4+y^4$

The coefficients form a triangular pyramid, with 1s on the edges, and each face a copy of Pascal's triangle.

Chapter 15 Binomial Theorem - Homework Answers

Lesson 15A | Binomial expansions and Pascal's triangle

Past-Paper Homework - Worked Solutions

PP 1 **The King's School, 2021, Year 12 Alternative Higher School Certificate Trial Examination, Mathematics Extension 1, Q2**

Take out the factor -1: $2x-1=-(1-2x)$, and $(-1)^3=-1$, so $(2x-1)^3=-(1-2x)^3$. Row 3 of Pascal's triangle is 1, 3, 3, 1, so

$$(1-2x)^3=1+3(-2x)+3(-2x)^2+(-2x)^3=1-6x+12x^2-8x^3.$$

Hence $(2x-1)^3=8x^3-12x^2+6x-1$. **Answer (B).** Option (A) has the wrong sign on the x^2 term, (C) has the wrong coefficients of x^2 and x, and (D) forgets that $(2x)^3=8x^3$.

PP 2 **Ascham School, 2020, Mathematics Extension 1 Trial HSC Examination, Q8**

Take out the factor -4: $3x-4=-4\left(1-\dfrac{3x}{4}\right)$, so $(3x-4)^6=(-4)^6\left(1-\dfrac{3x}{4}\right)^6=4096\left(1-\dfrac{3x}{4}\right)^6$.

Row 6 is 1, 6, 15, 20, 15, 6, 1. The expansion has 7 terms, so the fourth term is the middle term, whichever end you count from. It is

$$4096\times 20\left(-\frac{3x}{4}\right)^3=4096\times 20\times\left(-\frac{27x^3}{64}\right)=-34560x^3.$$

The coefficient of the fourth term is -34560. **Answer (B).**

PP 3 **North Sydney Boys High School, 2020, Year 12 Assessment Task 3, Mathematics Extension 1, Q2**

Take out the factor x: $(a+x)^6=(x+a)^6=x^6\left(1+\dfrac{a}{x}\right)^6$. Row 6 is 1, 6, 15, 20, 15, 6, 1. Since $x^3=x^6\times x^{-3}$, we need the term of the bracket in x^{-3}, which is $20\left(\dfrac{a}{x}\right)^3=\dfrac{20a^3}{x^3}$. So the term in x^3 is $x^6\times\dfrac{20a^3}{x^3}=20a^3x^3$.

Then $20a^3=1280$, so $a^3=64$ and $a=4$. **Answer (B).**

PP 4 **Girraween High School, 2021, Trial Higher School Certificate Examination, Mathematics Extension 1, Q2**

Take out the factor $2x$: $(2x+k)^6=(2x)^6\left(1+\dfrac{k}{2x}\right)^6=64x^6\left(1+\dfrac{k}{2x}\right)^6$. Row 6 is 1, 6, 15, 20, 15, 6, 1.

Since $x=x^6\times x^{-5}$, the term in x comes from $6\left(\dfrac{k}{2x}\right)^5=\dfrac{6k^5}{32x^5}$, so it is $64x^6\times\dfrac{6k^5}{32x^5}=12k^5x$.

Since $x^2=x^6\times x^{-4}$, the term in x^2 comes from $15\left(\dfrac{k}{2x}\right)^4=\dfrac{15k^4}{16x^4}$, so it is $64x^6\times\dfrac{15k^4}{16x^4}=60k^4x^2$.

Putting $12k^5=60k^4$ gives $12k^4(k-5)=0$, so $k=0$ or $k=5$. None of the options is 0, so $k=5$. **Answer (A).**

PP 5 **Moriah College, 2023, Trial Higher School Certificate Examination, Mathematics Extension 1, Q9(d)**

(i) Row 3 of Pascal's triangle is 1, 3, 3, 1, so $(1+x)^3=1+3x+3x^2+x^3$.

(ii) First $(2x-1)^2=4x^2-4x+1$, so

$$(2x-1)^2(1+x)^3=\left(4x^2-4x+1\right)\left(1+3x+3x^2+x^3\right).$$

The terms in x^4 come from $4x^2\times 3x^2=12x^4$ and $(-4x)\times x^3=-4x^4$. The constant 1 cannot give x^4, because $(1+x)^3$ stops at x^3. So the coefficient of x^4 is $12-4=8$.

PP 6 **Sydney Grammar School, 2023, Trial Examination, Form VI Mathematics Extension 1, Q13(a)**

Take out the factor x: $x-\dfrac{2}{x}=x\left(1-\dfrac{2}{x^2}\right)$, so $\left(x-\dfrac{2}{x}\right)^4=x^4\left(1-\dfrac{2}{x^2}\right)^4$. Row 4 is 1, 4, 6, 4, 1, so

$$\left(1-\frac{2}{x^2}\right)^4=1+4\left(-\frac{2}{x^2}\right)+6\left(-\frac{2}{x^2}\right)^2+4\left(-\frac{2}{x^2}\right)^3+\left(-\frac{2}{x^2}\right)^4=1-\frac{8}{x^2}+\frac{24}{x^4}-\frac{32}{x^6}+\frac{16}{x^8}.$$

Multiplying by x^4,

$$\left(x-\frac{2}{x}\right)^4=x^4-8x^2+24-\frac{32}{x^2}+\frac{16}{x^4}.$$

PP 7 **Girraween High School, 2023, Year 12 Trial Examination, Mathematics Extension 1, Q16(a)**

Take out the factor x: $x+\frac{2}{x}=x\left(1+\frac{2}{x^2}\right)$, so $\left(x+\frac{2}{x}\right)^6=x^6\left(1+\frac{2}{x^2}\right)^6$.

Row 6 is 1, 6, 15, 20, 15, 6, 1. The constant term needs the term of the bracket in x^{-6}. This is the fourth term, $20\left(\frac{2}{x^2}\right)^3=\frac{160}{x^6}$.

Multiplying by x^6, the constant term is 160.

PP 8 **Abbotsleigh, 2021, Higher School Certificate Assessment Task 4, Mathematics Extension 1, Q12(c)**

Take out the factor $3x^2$: $3x^2-2a=3x^2\left(1-\frac{2a}{3x^2}\right)$, so

$$\left(3x^2-2a\right)^5=\left(3x^2\right)^5\left(1-\frac{2a}{3x^2}\right)^5=243x^{10}\left(1-\frac{2a}{3x^2}\right)^5.$$

Row 5 is 1, 5, 10, 10, 5, 1. Since $x^6=x^{10}\times x^{-4}$, we need the term of the bracket in x^{-4}. That is the third term, $10\left(-\frac{2a}{3x^2}\right)^2=\frac{40a^2}{9x^4}$. So the term in x^6 is

$$243x^{10}\times\frac{40a^2}{9x^4}=1080a^2x^6,$$

and the coefficient of x^6 is $1080a^2$.

PP 9 **Normanhurst Boys High School, 2022, Year 12 Course Assessment Task 4 (Trial Examination), Mathematics Extension 1, Q11(b)**

Take out the factor $3x$: $3x-\frac{1}{x^2}=3x\left(1-\frac{1}{3x^3}\right)$, so $\left(3x-\frac{1}{x^2}\right)^6=729x^6\left(1-\frac{1}{3x^3}\right)^6$.

Row 6 is 1, 6, 15, 20, 15, 6, 1. The constant term needs the term of the bracket in x^{-6}. This is the third term, $15\left(-\frac{1}{3x^3}\right)^2=\frac{15}{9x^6}$.

Multiplying by $729x^6$, the constant term is

$$729\times\frac{15}{9}=1215=\binom{6}{2}3^4.$$

Chapter 15 Binomial Theorem - Homework Answers

Lesson 15B | Further binomial expansions

Lesson Examples

C1 **Tutor demo.**

a $(3-2x)^4 = 81 - 216x + 216x^2 - 96x^3 + 16x^4$

b $\left(2x + \frac{1}{2}a\right)^5 = 32x^5 + 40ax^4 + 20a^2x^3 + 5a^3x^2 + \frac{5}{8}a^4x + \frac{1}{32}a^5$

Your turn.

a $(2-5x)^3 = 8 - 60x + 150x^2 - 125x^3$

b $\left(3x + \frac{1}{3}a\right)^4 = 81x^4 + 36ax^3 + 6a^2x^2 + \frac{4}{9}a^3x + \frac{1}{81}a^4$

C2 **Tutor demo.**

$\left(x + 3x^{-2}\right)^6 = x^6 + 6x^5\left(3x^{-2}\right) + 15x^4\left(3x^{-2}\right)^2 + 20x^3\left(3x^{-2}\right)^3 + 15x^2\left(3x^{-2}\right)^4 + 6x\left(3x^{-2}\right)^5 + \left(3x^{-2}\right)^6$

a $15 \times x^4 \times 9x^{-4} = 135$ b $20 \times x^3 \times 27x^{-6} = 540x^{-3}$

Your turn. $\left(x + 2x^{-1}\right)^4 = x^4 + 4x^3\left(2x^{-1}\right) + 6x^2\left(2x^{-1}\right)^2 + 4x\left(2x^{-1}\right)^3 + \left(2x^{-1}\right)^4$

a $6 \times x^2 \times 4x^{-2} = 24$ b $4 \times x \times 8x^{-3} = 32x^{-2}$

C3 **Tutor demo.** $(3-2x)^6 = 729 - 2916x + 4860x^2 - \cdots$, so the term in x^2 is $4 \times 4860x^2 - x \times 2916x = 16524x^2$

Your turn. $(2-x)^5 = 32 - 80x + 80x^2 - \cdots$, so the term in x^2 is $3 \times 80x^2 - x \times 80x = 160x^2$

Textbook Homework

1

a $x^4 + 4x^3y + 6x^2y^2 + 4xy^3 + y^4$

b $x^4 - 4x^3y + 6x^2y^2 - 4xy^3 + y^4$

c $r^6 - 6r^5s + 15r^4s^2 - 20r^3s^3 + 15r^2s^4 - 6rs^5 + s^6$

d $p^{10} + 10p^9q + 45p^8q^2 + 120p^7q^3 + 210p^6q^4 + 252p^5q^5 + 210p^4q^6 + 120p^3q^7 + 45p^2q^8 + 10pq^9 + q^{10}$

e $a^9 - 9a^8b + 36a^7b^2 - 84a^6b^3 + 126a^5b^4 - 126a^4b^5 + 84a^3b^6 - 36a^2b^7 + 9ab^8 - b^9$

f $32x^5 + 80x^4y + 80x^3y^2 + 40x^2y^3 + 10xy^4 + y^5$

g $p^7 - 14p^6q + 84p^5q^2 - 280p^4q^3 + 560p^3q^4 - 672p^2q^5 + 448pq^6 - 128q^7$

h $81x^4 + 216x^3y + 216x^2y^2 + 96xy^3 + 16y^4$

i $a^3 - \dfrac{3a^2b}{2} + \dfrac{3ab^2}{4} - \dfrac{b^3}{8}$

j $\dfrac{r^5}{32} + \dfrac{5r^4s}{48} + \dfrac{5r^3s^2}{36} + \dfrac{5r^2s^3}{54} + \dfrac{5rs^4}{162} + \dfrac{s^5}{243}$

k $x^6 + 6x^4 + 15x^2 + 20 + \dfrac{15}{x^2} + \dfrac{6}{x^4} + \dfrac{1}{x^6}$

2

a $1 + 4x^2 + 6x^4 + 4x^6 + x^8$

b $1 - 9x^2 + 27x^4 - 27x^6$

c $x^{12} + 12x^{10}y^3 + 60x^8y^6 + 160x^6y^9 + 240x^4y^{12} + 192x^2y^{15} + 64y^{18}$

d $x^9 - 9x^7 + 36x^5 - 84x^3 + 126x - \dfrac{126}{x} + \dfrac{84}{x^3} - \dfrac{36}{x^5} + \dfrac{9}{x^7} - \dfrac{1}{x^9}$

e $x^3\sqrt{x} + 7x^3\sqrt{y} + 21x^2y\sqrt{x} + 35x^2y\sqrt{y} + 35xy^2\sqrt{x} + 21xy^2\sqrt{y} + 7y^3\sqrt{x} + y^3\sqrt{y}$

f $\dfrac{32}{x^5} + \dfrac{240}{x^2} + 720x + 1080x^4 + 810x^7 + 243x^{10}$

3

a $\left(y + (x-y)\right)^5 = x^5$ b $\left(a - (a-b)\right)^4 = b^4$

c $\left(x + (2y-x)\right)^3 = (2y)^3 = 8y^3$ d $\left((x+y) - (x-y)\right)^6 = (2y)^6 = 64y^6$

4

a (i) $1024 + 1280x + 640x^2 + 160x^3 + \cdots$ (ii) -160

b (i) $1 - 12x + 60x^2 - 160x^3 + 240x^4 - \cdots$ (ii) 720

c (i) $2187 - 5103y + 5103y^2 - 2835y^3 + 945y^4 - \cdots$ (ii) 11718

5

a $2x^6 + 30x^4y^2 + 30x^2y^4 + 2y^6$

6

a 540 b 48 c -960 d -8

7

a (i) $x^3 + 3x^2h + 3xh^2 + h^3$ (ii) $3x^2h + 3xh^2 + h^3$ (iii) $3x^2$ b $5x^4$

8 **b** 466 **c** 42

9 $\frac{7}{2}$

10 **a** 1.10408 **b** 0.90392

11 **a** (i) $\left(x^3+\frac{1}{x^3}\right)+3\left(x+\frac{1}{x}\right)$

(ii) $\left(x^5+\frac{1}{x^5}\right)+5\left(x^3+\frac{1}{x^3}\right)+10\left(x+\frac{1}{x}\right)$

(iii) $\left(x^7+\frac{1}{x^7}\right)+7\left(x^5+\frac{1}{x^5}\right)+21\left(x^3+\frac{1}{x^3}\right)+35\left(x+\frac{1}{x}\right)$

b (i) 2 (ii) 2 (iii) 2

12 $a=3$ or $a=-3$

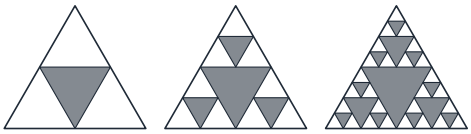
14 **a** $x^6+6x^4+15x^2+20+\frac{15}{x^2}+\frac{6}{x^4}+\frac{1}{x^6}$

b $A=-6$, $B=9$ and $C=-2$.

15 **a** $x^3+y^3+z^3+6xyz+3x^2y+3xy^2+3xz^2+3x^2z+3y^2z+3yz^2$

b 19

16 **a** The limiting figure for this process is called the Sierpinski Gasket. It is one of the classic regular fractals.



b Sierpinski's triangle is formed.

Chapter 15 Binomial Theorem - Homework Answers

Lesson 15B | Further binomial expansions

Past-Paper Homework - Worked Solutions

PP 1 **Sydney Grammar School, 2021, Trial Examination, Form VI Mathematics Extension 1, Q2**

Row 6 of Pascal's triangle is 1, 6, 15, 20, 15, 6, 1. In $(2x+5)^6$ the powers of $2x$ go down from 6 while the powers of 5 go up from 0. The term in x^2 contains $(2x)^2$, so it is the fifth term:

$$15 \times (2x)^2 \times 5^4 = 15 \times 4x^2 \times 625 = 37500x^2.$$

The coefficient is 37500. **Answer (C).** Option (A) is only the number from the triangle, and (D) leaves out the 15.

PP 2 **Hurlstone Agricultural High School, 2021, Higher School Certificate Assessment Task 4, Mathematics Extension 1, Q2**

Row 6 is 1, 6, 15, 20, 15, 6, 1, so

$$\left(x-\sqrt{2}\right)^6 = x^6 + 6x^5\left(-\sqrt{2}\right) + 15x^4\left(-\sqrt{2}\right)^2 + \cdots$$

The term in x^4 is $15x^4 \times 2 = 30x^4$, because $\left(-\sqrt{2}\right)^2 = 2$ is positive. The coefficient is 30. **Answer (B).** Option (A) has the wrong sign.

PP 3 **Hurlstone Agricultural High School, 2023, Higher School Certificate Trial Examination, Mathematics Extension 1, Q9**

Write $\left(2x^2+\dfrac{1}{x}\right)^5 = \left(2x^2+x^{-1}\right)^5$. Row 5 is 1, 5, 10, 10, 5, 1. The first term $\left(2x^2\right)^5$ has x^{10}. From one term to the next, one factor x^2 is replaced by x^{-1}, so the power of x goes down by 3: $x^{10},\ x^7,\ x^4,\ x,\ \ldots$ The term in x is the fourth term:

$$10 \times \left(2x^2\right)^2 \times \left(x^{-1}\right)^3 = 10 \times 4x^4 \times x^{-3} = 40x.$$

The coefficient is 40. **Answer (B).**

PP 4 **Sydney Girls High School, 2023, Trial Higher School Certificate Examination, Mathematics Extension 1, Q11(c)**

Write $\left(x-\dfrac{2}{x^2}\right)^6 = \left(x-2x^{-2}\right)^6$. Row 6 is 1, 6, 15, 20, 15, 6, 1, so

$$\left(x-2x^{-2}\right)^6 = x^6 + 6x^5\left(-2x^{-2}\right) + 15x^4\left(-2x^{-2}\right)^2 + 20x^3\left(-2x^{-2}\right)^3 + \cdots$$

The powers of x are $x^6,\ x^3,\ x^0,\ \ldots$, going down by 3. So the constant term is the third term:

$$15 \times x^4 \times 4x^{-4} = 60.$$

PP 5 **Knox Grammar School, 2023, Higher School Certificate Trial Examination, Mathematics Extension 1, Q11(a)**

Write $\left(x+\dfrac{2}{x}\right)^{10} = \left(x+2x^{-1}\right)^{10}$. Row 10 is 1, 10, 45, 120, 210, 252, 210, 120, 45, 10, 1. The powers of x are $x^{10},\ x^8,\ x^6,\ x^4,\ \ldots$, going down by 2, so the term in x^4 is the fourth term:

$$120 \times x^7 \times \left(2x^{-1}\right)^3 = 120 \times x^7 \times 8x^{-3} = 960x^4.$$

The coefficient of x^4 is 960.

PP 6 **Ascham School, 2023, Year 12 Trial Examination, Mathematics Extension 1, Q12(b)**

Row 5 is 1, 5, 10, 10, 5, 1. Expand only as far as the term in x^2:

$$\begin{aligned}(3-2x)^5 &= 3^5 + 5 \times 3^4 \times (-2x) + 10 \times 3^3 \times (-2x)^2 + \cdots \\ &= 243 - 810x + 1080x^2 - \cdots\end{aligned}$$

In $\left(2+x^2\right)(3-2x)^5$, the terms in x^2 are $2 \times 1080x^2$ and $x^2 \times 243$. So the coefficient of x^2 is

$$2 \times 1080 + 243 = 2403.$$

PP 7 **Pymble Ladies' College, 2023, Mathematics Extension 1 HSC Trial Examination Term 3, Q13(a)**

Write $\left(2x+\dfrac{5}{x^2}\right)^6=\left(2x+5x^{-2}\right)^6$. Row 6 is 1, 6, 15, 20, 15, 6, 1, so

$$\begin{aligned}\left(2x+5x^{-2}\right)^6&=(2x)^6+6(2x)^5\left(5x^{-2}\right)+15(2x)^4\left(5x^{-2}\right)^2+\cdots\\&=64x^6+960x^3+6000+\cdots\end{aligned}$$

The later terms have negative powers of x. The constant 2 needs the constant term 6000, and $-\dfrac{3}{x^3}=-3x^{-3}$ needs the term $960x^3$. So the term independent of x is

$$2\times 6000-3x^{-3}\times 960x^3=12000-2880=9120.$$

PP 8 **St Aloysius' College, 2024, Trial Higher School Certificate Examination, Mathematics Extension 1, Q14(c)**

Row 12 is 1, 12, 66, 220, 495, 792, 924, 792, 495, 220, 66, 12, 1. In $(a-bx)^{12}$ the powers of a go down from 12 while the powers of $-bx$ go up from 0. So the terms in x^7 and x^8 are the eighth and ninth terms:

$$792a^5(-bx)^7=-792a^5b^7x^7\quad\text{and}\quad 495a^4(-bx)^8=495a^4b^8x^8.$$

In $(1+x)(a-bx)^{12}$, the term in x^8 comes from $1\times 495a^4b^8x^8$ and $x\times\left(-792a^5b^7x^7\right)$. So the coefficient of x^8 is

$$495a^4b^8-792a^5b^7=99a^4b^7(5b-8a).$$

This is zero and $a,b\neq 0$, so $8a=5b$. Hence $\dfrac{a}{b}=\dfrac{5}{8}$.

PP 9 **James Ruse Agricultural High School, 2023, Year 12 Trial HSC Examination, Mathematics Extension 1, Q15(b)**

Write $\left(\dfrac{x^3}{2}+\dfrac{a}{x}\right)^8=\left(\dfrac{x^3}{2}+ax^{-1}\right)^8$. Row 8 is 1, 8, 28, 56, 70, 56, 28, 8, 1. The first term $\left(\dfrac{x^3}{2}\right)^8$ has x^{24}. From one term to the next, one factor x^3 is replaced by x^{-1}, so the power of x goes down by 4. The power x^0 is reached after 6 steps, in the seventh term:

$$28\times\left(\frac{x^3}{2}\right)^2\times\left(ax^{-1}\right)^6=28\times\frac{x^6}{4}\times a^6x^{-6}=7a^6.$$

Then $7a^6=5103$, so $a^6=729=3^6$, and $a=3$ or $a=-3$.

Chapter 15 Binomial Theorem - Homework Answers

Lesson 15C | The binomial theorem

Lesson Examples

C1 **Tutor demo.**

a $^{13}C_9 = \dfrac{13 \times 12 \times 11 \times 10}{4 \times 3 \times 2 \times 1} = 715$

b $^9C_5\, x^4y^5 = 126x^4y^5$

c $^8C_4 \times 3^4 \times 2^4 = 90720 = 2^5 \times 3^4 \times 5 \times 7$

d $^nC_3\, a^{n-3}(-3b)^3 = -\dfrac{9n(n-1)(n-2)}{2}\, a^{n-3}b^3$

Your turn.

a $^{10}C_6 = 210$

b $^8C_3\, x^5y^3 = 56x^5y^3$

c $-^6C_3 \times 2^3 \times 5^3 = -20000 = -2^5 \times 5^4$

d $^nC_2\, a^{n-2}(4b)^2 = 8n(n-1)\, a^{n-2}b^2$

C2 **Tutor demo.**

a $n^2 - n - 132 = 0$, so $(n-12)(n+11) = 0$ and $n = 12$

b $n^2 + n - 72 = 0$, so $(n-8)(n+9) = 0$ and $n = 8$

Your turn.

a $n^2 - n - 56 = 0$, so $(n-8)(n+7) = 0$ and $n = 8$

b $n^2 + n - 30 = 0$, so $(n-5)(n+6) = 0$ and $n = 5$

C3 **Tutor demo.** In $\left(x^3 - 2x^{-1}\right)^{12}$ the powers of x go down by 4 from x^{36}.

a x^0 after 9 steps: $^{12}C_9\left(x^3\right)^3\left(-2x^{-1}\right)^9 = 220 \times (-512) = -112640$

b x^8 after 7 steps: $^{12}C_7\left(x^3\right)^5\left(-2x^{-1}\right)^7 = 792 \times (-128)x^8 = -101376x^8$

Your turn. In $\left(x + 2x^{-2}\right)^9$ the powers of x go down by 3 from x^9.

a x^0 after 3 steps: $^9C_3\, x^6\left(2x^{-2}\right)^3 = 84 \times 8 = 672$

b x^{-6} after 5 steps: $^9C_5\, x^4\left(2x^{-2}\right)^5 = 126 \times 32x^{-6} = 4032x^{-6}$

Textbook Homework

1 a 4 b 20 c 9 d 35

2 a 10 b 1 c 11 d 210

3 a 3003 b 1287 c 792 d 22 e 429 f 1292

4 a $^4C_0 = 1$, $^4C_1 = 4$, $^4C_2 = 6$, $^4C_3 = 4$, $^4C_4 = 1$

b (i) 16 (ii) 0

5 a 32 b 32 c 20 d 252

6 a (i) both are 56 (ii) both are 35

b $n = 5$ c $n = 4$ or $n = 8$

7 b (i) $n = 6$ (ii) $n = 10$

8 a $8568x^5$ b $2217093120x^9$ c $-19208x^3$ d 189

9 a $672x^2$ b $\dfrac{1001x^9y^5}{16}$ c $-\dfrac{33x^{10}y^2}{1024}$ d $190a^2b^9$

10 a $x = \frac{11}{2}$ b $x = -\frac{7}{3}$

11 a $5x^2 : 39$ b $5 : 2$ c $18304 : 1$

12 a (i) 1 (ii) n (iii) $\frac{1}{2}n(n-1)$ (iv) $\frac{1}{6}n(n-1)(n-2)$

b (i) 16 (ii) 9 (iii) 4 (iv) 6 (v) 4 (vi) 7

13 a $a = 2$ and $n = 14$ b $a = -\frac{1}{3}$ and $n = 10$

14 a $n = 14$ b $n = 13$

15 ${}^{40}C_{20} \approx 1.378 \times 10^{11}$

17 **a** ${}^{n}C_0\, x^n + {}^{n}C_1\, x^{n-1}h + {}^{n}C_2\, x^{n-2}h^2 + \cdots + {}^{n}C_n\, h^n$
b nx^{n-1}

18 The second member of row n is ${}^{n}C_1 = n$. Suppose that n is prime, and let $2 \leq r \leq n-1$. Then ${}^{n}C_r \times r!\,(n-r)! = n! = n \times (n-1)!$, so n divides ${}^{n}C_r \times r!\,(n-r)!$. Every factor of $r!$ and of $(n-r)!$ is less than n, so the prime n has no common factor with $r!\,(n-r)!$. Hence n divides ${}^{n}C_r$. Also ${}^{n}C_1 = {}^{n}C_{n-1} = n$. So every number in row n, excluding the 1s at the ends, is divisible by n.

19 **a** 3 points, 3 segments, 1 triangle
b 4 points, 6 segments, 4 triangles, 1 quadrilateral
c 5 points, 10 segments, 10 triangles, 5 quadrilaterals, 1 pentagon
d 21

20 **b** (i) $1 + x + x^2 + x^3 + \cdots$
(ii) $1 + 2x + 3x^2 + 4x^3 + \cdots$
(iii) $1 - 2x + 3x^2 - 4x^3 + \cdots$
(iv) $1 + \frac{1}{2}x - \frac{1}{8}x^2 + \frac{1}{16}x^3 - \cdots$
c Using part b(i),

$$\begin{aligned}\text{LHS} &= \frac{d}{dx}\left(\frac{1}{1-x}\right) \\ &= \frac{d}{dx}\left(1 + x + x^2 + x^3 + \cdots\right) \\ &= 1 + 2x + 3x^2 + \cdots \\ &= \text{RHS by part b(ii).}\end{aligned}$$

Chapter 15 Binomial Theorem - Homework Answers

Lesson 15C | The binomial theorem

Past-Paper Homework - Worked Solutions

PP 1 **Sydney Grammar School, 2020, Trial Examination, Form VI Mathematics Extension 1, Q2**

By the binomial theorem, the term in x^3 has $(-3x)^3$, so

$$(2-3x)^8 = \cdots + {}^8C_3 \times 2^5 \times (-3x)^3 + \cdots$$

Here ${}^8C_3 = \dfrac{8\times 7\times 6}{3\times 2\times 1} = 56$, so the term is

$$56 \times 32 \times (-27x^3) = -48384x^3.$$

The coefficient is -48384. **Answer (D).** Option (A) is only ${}^8C_3$, and (B) leaves out the factor 2^5.

PP 2 **Sydney Grammar School, 2022, Trial Examination, Form VI Mathematics Extension 1, Q7**

Write $\left(4x^2 - \dfrac{1}{x}\right)^9 = \left(4x^2 - x^{-1}\right)^9$. The first term $\left(4x^2\right)^9$ has x^{18}. From one term to the next, one factor x^2 is replaced by x^{-1}, so the power of x goes down by 3. The power x^0 is reached after 6 steps, so the constant term is

$${}^9C_6 \times \left(4x^2\right)^3 \times \left(-x^{-1}\right)^6 = 84 \times 64x^6 \times x^{-6} = 5376.$$

Answer (D). Option (C) is only ${}^9C_6$, and (B) has the wrong sign: $\left(-x^{-1}\right)^6$ is positive.

PP 3 **Abbotsleigh, 2023, Higher School Certificate Assessment 4 Trial Examination, Mathematics Extension 1, Q11(c)**

By the binomial theorem, the term in x^6 has $(2x)^6$ and $(-p)^3$:

$$(2x-p)^9 = \cdots + {}^9C_3 \times (2x)^6 \times (-p)^3 + \cdots$$

So the coefficient of x^6 is $84 \times 64 \times (-p^3) = -5376p^3$. Then

$$-5376p^3 = -672000, \quad \text{so} \quad p^3 = 125 \quad \text{and} \quad p = 5.$$

PP 4 **North Sydney Girls High School, 2023, HSC Trial Examination, Mathematics Extension 1, Q12(b)**

(i) By the binomial theorem,

$$(2+3x)^6 = \cdots + {}^6C_k \times 2^{6-k} \times (3x)^k + \cdots$$

so the coefficient of x^k is ${}^6C_k \times 2^{6-k} \times 3^k$.

(ii) In $(2+3x)^6(1-4x^2)$, the term in x^3 comes from $1 \times (\text{term in } x^3)$ and $-4x^2 \times (\text{term in } x)$. By part (i), the coefficients of x^3 and x in $(2+3x)^6$ are

$${}^6C_3 \times 2^3 \times 3^3 = 4320 \quad \text{and} \quad {}^6C_1 \times 2^5 \times 3 = 576.$$

So the coefficient of x^3 is

$$4320 - 4 \times 576 = 2016.$$

PP 5 **North Sydney Girls High School, 2020, HSC Trial Examination, Mathematics Extension 1, Q11(e)**

Write $\left(2x^2 - \dfrac{3}{x}\right)^9 = \left(2x^2 - 3x^{-1}\right)^9$. The first term $\left(2x^2\right)^9$ has x^{18}. From one term to the next, one factor x^2 is replaced by x^{-1}, so the power of x goes down by 3. The power x^0 is reached after 6 steps, so the term independent of x is

$${}^9C_6 \times \left(2x^2\right)^3 \times \left(-3x^{-1}\right)^6 = 84 \times 8x^6 \times 729x^{-6} = 489888.$$

PP 6 **North Sydney Boys High School, 2020, Year 12 Assessment Task 3, Mathematics Extension 1, Q11(b)**

Write $\left(x^2 - \dfrac{1}{x^2}\right)^{10} = \left(x^2 - x^{-2}\right)^{10}$. The first term $\left(x^2\right)^{10}$ has x^{20}. From one term to the next, one factor x^2 is replaced by $-x^{-2}$, so the power of x goes down by 4. The power x^0 is reached after 5 steps, so the term independent of x is

$${}^{10}C_5 \times \left(x^2\right)^5 \times \left(-x^{-2}\right)^5 = 252 \times x^{10} \times \left(-x^{-10}\right) = -252.$$

PP 7 **Hornsby Girls High School, 2023, Year 12 Higher School Certificate Trial Examination, Mathematics Extension 1, Q11(d)**

By the binomial theorem, the coefficients of x^6 and x^8 in $(1+2x)^n$ are ${}^nC_6 \times 2^6$ and ${}^nC_8 \times 2^8$. So

$$\frac{{}^nC_6 \times 2^6}{{}^nC_8 \times 2^8} = \frac{7}{3}.$$

Now

$$\frac{{}^nC_6}{{}^nC_8} = \frac{n!}{6!\,(n-6)!} \times \frac{8!\,(n-8)!}{n!} = \frac{8 \times 7}{(n-6)(n-7)},$$

so

$$\begin{aligned} \frac{56}{4(n-6)(n-7)} &= \frac{7}{3} \\ (n-6)(n-7) &= 6 \\ n^2 - 13n + 36 &= 0 \\ (n-4)(n-9) &= 0 \end{aligned}$$

There is a term in x^8, so $n \geq 8$. Hence $n = 9$.

PP 8 **Normanhurst Boys High School, 2021, Year 12 Course Assessment Task 4 (Trial Examination), Mathematics Extension 1, Q11(d)**

Write $\left(px^3 - \frac{q}{x^2}\right)^9 = \left(px^3 - qx^{-2}\right)^9$. The first term is $\left(px^3\right)^9 = p^9x^{27}$, so the coefficient of x^{27} is p^9. From one term to the next, one factor x^3 is replaced by x^{-2}, so the power of x goes down by 5: $x^{27},\ x^{22},\ x^{17},\ \ldots$ The term in x^{17} is reached after 2 steps:

$${}^9C_2 \times \left(px^3\right)^7 \times \left(-qx^{-2}\right)^2 = 36p^7q^2x^{17}.$$

The coefficients are equal, so $p^9 = 36p^7q^2$. Since $p \neq 0$, $p^2 = 36q^2$. Since p and q are positive, $p = 6q$, that is, $p - 6q = 0$.

Chapter 15 Binomial Theorem - Homework Answers

Lesson 15D | Identities in Pascal's triangle

Lesson Examples

C1 **Tutor demo.**

a Substitute $x = 3$ into $(1+x)^n = {}^nC_0 + {}^nC_1\,x + {}^nC_2\,x^2 + \cdots + {}^nC_n\,x^n$. Because $1+3=4$, $4^n = {}^nC_0 + 3\,{}^nC_1 + 9\,{}^nC_2 + \cdots + 3^n\,{}^nC_n$.

b $1 + 3\times 4 + 9\times 6 + 27\times 4 + 81\times 1 = 1 + 12 + 54 + 108 + 81 = 256 = 4^4$

c Choosing 3 of the 8 elements to be in a subset leaves the other 5 out. So each 3-member subset pairs with exactly one 5-member subset, and ${}^8C_3 = {}^8C_5 = 56$.

Your turn.

a Substitute $x = 4$ into the same expansion. Because $1+4=5$, $5^n = {}^nC_0 + 4\,{}^nC_1 + 16\,{}^nC_2 + \cdots + 4^n\,{}^nC_n$.

b $1 + 4\times 3 + 16\times 3 + 64\times 1 = 1 + 12 + 48 + 64 = 125 = 5^3$

c Choosing 2 of the 9 elements to be in a subset leaves the other 7 out. So each 2-member subset pairs with exactly one 7-member subset, and ${}^9C_2 = {}^9C_7 = 36$.

C2 **Tutor demo.** $5(1+x)^4 = {}^5C_1 + 2\,{}^5C_2\,x + 3\,{}^5C_3\,x^2 + 4\,{}^5C_4\,x^3 + 5\,{}^5C_5\,x^4$

a $x = 1$: $5\times 2^4 = 80$. Check: $5 + 2\times 10 + 3\times 10 + 4\times 5 + 5\times 1 = 80$.

b $x = -1$: $5\times 0^4 = 0$. Check: $5 - 2\times 10 + 3\times 10 - 4\times 5 + 5\times 1 = 0$.

Your turn. $3(1+x)^2 = {}^3C_1 + 2\,{}^3C_2\,x + 3\,{}^3C_3\,x^2$

a $x = 1$: $3\times 2^2 = 12$. Check: $3 + 2\times 3 + 3\times 1 = 12$.

b $x = -1$: $3\times 0^2 = 0$. Check: $3 - 2\times 3 + 3\times 1 = 0$.

C3 **Tutor demo.**

a $r\times\dfrac{n!}{r!\,(n-r)!} = \dfrac{n!}{(r-1)!\,(n-r)!} = n\times\dfrac{(n-1)!}{(r-1)!\,(n-r)!} = n\times {}^{n-1}C_{r-1}$, because $(n-1)-(r-1) = n-r$.

b $(1+x)^2 = 1 + 2x + x^2$. On the left, the coefficient of x^2 is $1\times {}^nC_2 + 2\times {}^nC_1 + 1\times {}^nC_0$. On the right it is ${}^{n+2}C_2$. Check: $3 + 6 + 1 = 10 = {}^5C_2$.

Your turn.

a $(n-r)\times\dfrac{n!}{r!\,(n-r)!} = \dfrac{n!}{r!\,(n-r-1)!} = n\times\dfrac{(n-1)!}{r!\,(n-1-r)!} = n\times {}^{n-1}C_r$

b On the left, the coefficient of x^3 is $1\times {}^nC_3 + 2\times {}^nC_2 + 1\times {}^nC_1$. On the right it is ${}^{n+2}C_3$. Check: $4 + 12 + 4 = 20 = {}^6C_3$.

Textbook Homework

1 **a** (i) $1 + 4 + 6 + 4 + 1 = 16 = 2^4$

(ii) The sum $1 + 6 + 1$ of the first, third and fifth terms on the row equals the sum $4 + 4$ of the second and fourth terms.

(iii) The sum of the first, third and fifth terms on the row is half the sum of the whole row.

b (i) $4(1+x)^3 = {}^4C_1 + 2\,{}^4C_2\,x + 3\,{}^4C_3\,x^2 + 4\,{}^4C_4\,x^3$

(ii) $1\times 4 + 2\times 6 + 3\times 4 + 4\times 1 = 32 = 4\times 2^3$

(iii) $1\times 4 - 2\times 6 + 3\times 4 - 4\times 1 = 0$

4 **a** There are 10: $\{A,B\}$, $\{A,C\}$, $\{A,D\}$, $\{A,E\}$, $\{B,C\}$, $\{B,D\}$, $\{B,E\}$, $\{C,D\}$, $\{C,E\}$, $\{D,E\}$.

b There are 10: $\{A,B,C\}$, $\{A,B,D\}$, $\{A,B,E\}$, $\{A,C,D\}$, $\{A,C,E\}$, $\{A,D,E\}$, $\{B,C,D\}$, $\{B,C,E\}$, $\{B,D,E\}$, $\{C,D,E\}$.

c C, D and E.

d Given 5 letters, choose 2. Those that are left form a set of 3. Thus for every set of 2, there is a corresponding set of 3. Thus ${}^5C_2 = {}^5C_3$.

e Given n people, choose r. Those that are left form a set of $n-r$. Thus for every set of r, there is a corresponding set of $n-r$. Thus ${}^nC_r = {}^nC_{n-r}$.

5 **a** The 16 subsets are:

$\varnothing$

$\{A\}, \{B\}, \{C\}, \{D\}$

$\{A, B\}, \{A, C\}, \{A, D\}, \{B, C\}, \{B, D\}, \{C, D\}$

$\{A, B, C\}, \{A, B, D\}, \{A, C, D\}, \{B, C, D\}$

$\{A, B, C, D\}$

To form a subset of S, take each of A, B, C and D in turn and decide whether it is in or out. Thus the total number of subsets of S is $2 \times 2 \times 2 \times 2 = 2^4$.

b The LHS is the sum of the numbers of 0-member, 1-member, 2-member, 3-member and 4-member subsets, and so is also the number of all subsets.

c Generalise the previous argument to an n-member set.

6 **a** (i) There are four of them: $\{A, B, C\}$, $\{A, B, D\}$, $\{A, C, D\}$, $\{B, C, D\}$.

b (i) There are six of them: $\{A, B, E\}$, $\{A, C, E\}$, $\{A, D, E\}$, $\{B, C, E\}$, $\{B, D, E\}$, $\{C, D, E\}$. (ii) Omit E from each set.

c Let T be a 3-letter subset of S. If T does not contain E, then T is one of the 3-letter subsets of U. If T does contain E, then remove E, and the remaining 2-letter subset pairs with one of the 2-letter subsets of U.

d Generalise the previous argument.

7 **a** The five sequences, and the value of both sides for each, are:

$1, 7, 21, 35$: both sides are $\frac{1}{2}$

$7, 21, 35, 35$: both sides are $\frac{3}{4}$

$21, 35, 35, 21$: both sides are 1

$35, 35, 21, 7$: both sides are $\frac{5}{4}$

$35, 21, 7, 1$: both sides are $\frac{3}{2}$

c Using the addition property, $\text{LHS} = \frac{{}^nC_{r-1}}{{}^{n+1}C_r} + \frac{{}^nC_{r+1}}{{}^{n+1}C_{r+2}}$ and $\text{RHS} = \frac{2 \times {}^nC_r}{{}^{n+1}C_{r+1}}$. Now use the formula for ${}^nC_r$:

$\frac{{}^nC_{r-1}}{{}^{n+1}C_r} = \frac{r}{n+1}$, $\frac{{}^nC_{r+1}}{{}^{n+1}C_{r+2}} = \frac{r+2}{n+1}$ and $\frac{{}^nC_r}{{}^{n+1}C_{r+1}} = \frac{r+1}{n+1}$, so both sides equal $\frac{2r+2}{n+1}$.

8 **a** As explained in Question 5, the LHS counts all the subsets of S. We can also count the subsets by choosing whether each element in turn goes into a subset or not, giving 2^n subsets.

b (i) First, every subset of S either contains A or does not contain A. Secondly, a subset containing A is paired up with the unique subset obtained by removing A from the subset.

(ii) Adding A to a subset without A changes the number of members from odd to even or from even to odd.

(iii) The LHS is the total number of even-order subsets, and the RHS is the total number of odd-order subsets. We have seen that they are paired with each other, so the LHS and RHS are equal.

9 **a** (i)

$$\begin{aligned} {}^7C_3 &= {}^6C_3 + {}^6C_2 = {}^5C_3 + {}^5C_2 + {}^6C_2 \\ &= {}^4C_3 + {}^4C_2 + {}^5C_2 + {}^6C_2 \\ &= {}^3C_3 + {}^3C_2 + {}^4C_2 + {}^5C_2 + {}^6C_2, \end{aligned}$$

and ${}^3C_3 = {}^2C_2 = 1$.

b (i) There are ${}^2C_2 = 1$ subsets with highest element 3. There are ${}^3C_2 = 3$ subsets with highest element 4. There are ${}^4C_2 = 6$ subsets with highest element 5. There are ${}^5C_2 = 10$ subsets with highest element 6. There are ${}^6C_2 = 15$ subsets with highest element 7. This makes 35 subsets with 3 members.

10 **a** A common denominator for the two fractions is required:

$$\begin{aligned}\text{LHS} &= \frac{n!}{r\times(r-1)!\times(n-r)!}+\frac{n!}{(r-1)!\times(n-r+1)\times(n-r)!}\\ &= \frac{(n-r+1)\times n!+r\times n!}{r\times(r-1)!\times(n-r+1)\times(n-r)!}\\ &= \frac{\big((n-r+1)+r\big)\times n!}{r!\times(n-r+1)!}\\ &= \frac{(n+1)\times n!}{r!\times(n-r+1)!}\\ &= \frac{(n+1)!}{r!\times(n-r+1)!}\\ &= \text{RHS}.\end{aligned}$$

b This is the addition property of Pascal's triangle.

12 **a** Regarding this as an $(a+b)$-letter word with a identical As and b identical Bs, the number of permutations is $\dfrac{(a+b)!}{a!\,b!} = {}^{a+b}C_a = {}^{a+b}C_b$.
b ${}^{2n}C_n$
c (i) ${}^nC_2$ (ii) ${}^nC_{n-2} = {}^nC_2$ (iii) ${}^5C_2\times{}^5C_2 = 100$

Chapter 15 Binomial Theorem - Homework Answers

Chapter 15 Review | Chapter review exercise

1
$$\begin{array}{ccccccc}&&&1&&&\\&&1&&1&&\\&1&&2&&1&\\1&&3&&3&&1\\1&&4&&6&&4&&1\end{array}$$

2 **a** $1+5x+10x^2+10x^3+5x^4+x^5$
b $1+10x+40x^2+80x^3+80x^4+32x^5$
c $1-9x+27x^2-27x^3$
d $1-4xy+6x^2y^2-4x^3y^3+x^4y^4$

3 **a** $1+35x+490x^2+\cdots$
b $(1-5x)\big(1+35x+490x^2+\cdots\big) = \cdots+(490-175)x^2+\cdots$. The coefficient of x^2 is 315.

4 **a** $1+7x+21x^2+35x^3+35x^4+21x^5+7x^6+x^7$
b $(1+0.02)^7 = 1+7\times0.02+21\times0.0004+\cdots = 1+0.14+0.0084+\cdots$, and the remaining terms are much smaller. The first decimal place will be 1.

5 **a** $81+216x+216x^2+96x^3+16x^4$
b $125-75x+15x^2-x^3$
c $32x^5+320x^4y+1280x^3y^2+2560x^2y^3+2560xy^4+1024y^5$
d $x^4-4x^2+6-\dfrac{4}{x^2}+\dfrac{1}{x^4}$

6 **a** 10 **b** 35 **c** 56 **d** 2

7 **a** 45 **b** 792 **c** 84 **d** 40 **e** $\dfrac{84}{5}$ **f** $\dfrac{286}{9}$

8 **a** 1 (first entry in a row)
b 0 (reversibility of rows)
c $2^5 = 32$ (sum of a row)
d 0 (addition formula)

9 $n = 8$

10 $(1-3x)(1+5x)^{14} = (1-3x)\big(\cdots+{}^{14}C_7\,(5x)^7+{}^{14}C_8\,(5x)^8+\cdots\big)$
The coefficient of x^8 will be ${}^{14}C_8\times5^8-3\times{}^{14}C_7\times5^7 = 368671875$.

11 **a** $(1+x)^n = \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \binom{n}{3}x^3 + \cdots$

b Substitute $x = 2$: $(1+2)^n = \binom{n}{0} + \binom{n}{1} \times 2 + \binom{n}{2} \times 4 + \binom{n}{3} \times 8 + \cdots$

c When $n = 4$, $\text{LHS} = 1 + 4 \times 2 + 6 \times 4 + 4 \times 8 + 1 \times 16 = 1 + 8 + 24 + 32 + 16 = 81 = 3^4$.

12 **b** It tells us that the sum of the row in Pascal's triangle indexed by n, that is, containing the coefficients of $(1+x)^n$, is 2^n.
c The list of all subsets of a set of n objects may be partitioned into sets of sizes $0, 1, 2, \ldots, n$. The sum of the number of sets of each type will be the total number of all subsets, which is 2^n.

13 **c** First identity: $1 \times 5 + 2 \times 10 + 3 \times 10 + 4 \times 5 + 5 \times 1 = 80 = 5 \times 2^4$.
Second identity: $1 \times 2 \times 10 + 2 \times 3 \times 10 + 3 \times 4 \times 5 + 4 \times 5 \times 1 = 160 = 4 \times 5 \times 2^3$.

Chapter 15 Binomial Theorem - Homework Answers

Lesson 15D | Identities in Pascal's triangle

Past-Paper Homework - Worked Solutions

PP 1 **North Sydney Boys High School, 2020, Year 12 Assessment Task 3, Mathematics Extension 1, Q14(a)**

By the binomial theorem,

$$(1+x)^n = \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \cdots + \binom{n}{n}x^n.$$

Substitute $x = 1$. Because $1 + 1 = 2$,

$$2^n = \binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \cdots + \binom{n}{n}.$$

PP 2 **Sydney Boys High School, 2020, Year 12 Term 3 Trial HSC, Mathematics Extension 1, Q14(c)(i)**

Differentiate both sides of $(1+x)^n = {}^nC_0 + {}^nC_1\,x + {}^nC_2\,x^2 + {}^nC_3\,x^3 + \cdots + {}^nC_n\,x^n$ with respect to x:

$$n(1+x)^{n-1} = {}^nC_1 + 2\,{}^nC_2\,x + 3\,{}^nC_3\,x^2 + \cdots + n\,{}^nC_n\,x^{n-1}.$$

Substitute $x = 1$:

$$n2^{n-1} = {}^nC_1 + 2\,{}^nC_2 + 3\,{}^nC_3 + \cdots + n\,{}^nC_n.$$

PP 3 **Killara High School, 2021, Year 12 Mathematics Extension 1 Trial Examination, Q13(e)**

By the binomial theorem,

$$(1+x)^{2n} = {}^{2n}C_0 + {}^{2n}C_1\,x + {}^{2n}C_2\,x^2 + {}^{2n}C_3\,x^3 + \cdots + {}^{2n}C_{2n}\,x^{2n},$$
$$(1-x)^{2n} = {}^{2n}C_0 - {}^{2n}C_1\,x + {}^{2n}C_2\,x^2 - {}^{2n}C_3\,x^3 + \cdots + {}^{2n}C_{2n}\,x^{2n}.$$

Adding, the odd powers of x cancel:

$$(1+x)^{2n} + (1-x)^{2n} = 2\left({}^{2n}C_0 + {}^{2n}C_2\,x^2 + {}^{2n}C_4\,x^4 + \cdots + {}^{2n}C_{2n}\,x^{2n}\right).$$

Now put $n = 50$ and $x = 1$. Because ${}^{100}C_0 = 1$,

$$2^{100} + 0^{100} = 2\left(1 + {}^{100}C_2 + {}^{100}C_4 + \cdots + {}^{100}C_{100}\right),$$

so

$$1 + {}^{100}C_2 + {}^{100}C_4 + \cdots + {}^{100}C_{100} = \frac{2^{100}}{2} = 2^{99}.$$

PP 4 **Blacktown Boys' High School, 2024, HSC Trial Examination, Mathematics Extension 1, Q12(d)**

The rows are reversible, and $2026 - 1924 = 102$, so ${}^{2026}C_{1924} = {}^{2026}C_{102}$. By the given identity with $n = 2025$ and $r = 101$,

$${}^{2024}C_{100} + {}^{2024}C_{101} = {}^{2025}C_{101}.$$

By the given identity again, with $n = 2026$ and $r = 102$,

$${}^{2026}C_{102} = {}^{2025}C_{101} + {}^{2025}C_{102}.$$

Hence

$$\begin{aligned} {}^{2026}C_{1924} - {}^{2024}C_{100} - {}^{2024}C_{101} &= {}^{2026}C_{102} - {}^{2025}C_{101} \\ &= {}^{2025}C_{102}. \end{aligned}$$

So one possible set is $p = 2025$ and $q = 102$. (Another is $p = 2025$ and $q = 1923$.)

PP 5 **Pymble Ladies' College, 2023, Mathematics Extension 1 HSC Trial Examination Term 3, Q14(b)**

The club has n males and n females. Choosing any 3 of the $2n$ members, there are ${}^{2n}C_3$ committees. Now count the same committees by the number of males on them:

Males	Females	Number of committees
3	0	${}^{n}C_3$
2	1	${}^{n}C_2 \times n$
1	2	$n \times {}^{n}C_2$
0	3	${}^{n}C_3$

Both counts give the number of all the committees, so

$$ {}^{2n}C_3 = 2\,{}^{n}C_3 + 2n\,{}^{n}C_2, \quad \text{that is,} \quad n\,{}^{n}C_2 + {}^{n}C_3 = \frac{1}{2}\,{}^{2n}C_3. $$

PP 6 **Kinross Wolaroi School, 2023, Year 12 Mathematics Extension 1 Trial HSC Examination, Q14(d)**

Because $(1+x)^3(1+x)^6 = (1+x)^9$, the coefficients of x^6 on both sides are equal. On the right, the coefficient of x^6 is $\binom{9}{6}$.

On the left,

$$ (1+x)^3(1+x)^6 = \left(\binom{3}{0} + \binom{3}{1}x + \binom{3}{2}x^2 + \binom{3}{3}x^3\right)\left(\binom{6}{0} + \binom{6}{1}x + \cdots + \binom{6}{6}x^6\right). $$

A term in x^6 comes from the term in x^k in the first bracket times the term in x^{6-k} in the second bracket, for $k = 0, 1, 2, 3$. So the coefficient of x^6 on the left is

$$ \binom{3}{0}\binom{6}{6} + \binom{3}{1}\binom{6}{5} + \binom{3}{2}\binom{6}{4} + \binom{3}{3}\binom{6}{3}. $$

Equating the coefficients of x^6 gives the result.

PP 7 **North Sydney Boys High School, 2023, Year 12, Assessment task 3, Term 2, Mathematics Extension 1, Q14(c)**

By the binomial theorem, $a_k = {}^{n}C_k$ and $b_k = {}^{n+1}C_k$. Expand the right-hand side of $(1+x)^{n+1} = (1+x)^n(1+x)$:

$$ \left(a_0 + a_1x + \cdots + a_nx^n\right)(1+x) = a_0 + (a_0+a_1)x + (a_1+a_2)x^2 + \cdots + a_nx^{n+1}. $$

Equating the coefficients of x^k, $a_{k-1} + a_k = b_k$ for $k = 1, 2, \ldots, n$. Also $a_0 = 1$. So the expression is

$$ \frac{b_1b_2\cdots b_n}{a_0a_1a_2\cdots a_n} = \frac{b_1}{a_1} \times \frac{b_2}{a_2} \times \cdots \times \frac{b_n}{a_n}. $$

By the formula for ${}^{n}C_r$, for $k = 1, 2, \ldots, n$,

$$ \frac{b_k}{a_k} = \frac{(n+1)!}{k!\,(n+1-k)!} \times \frac{k!\,(n-k)!}{n!} = \frac{n+1}{n+1-k}. $$

So the expression is

$$ \frac{n+1}{n} \times \frac{n+1}{n-1} \times \cdots \times \frac{n+1}{1} = \frac{(n+1)^n}{n!}. $$